

Boston Scientific
Advancing science for life™

MAMBA™ Microcatheter Family

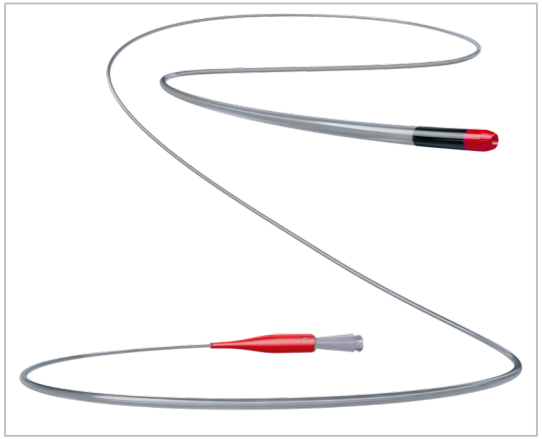
In-Service

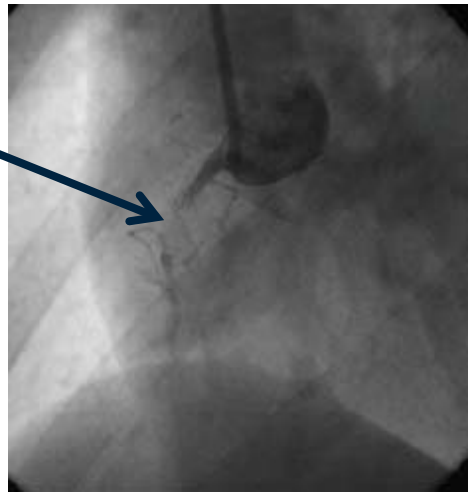
Complex PCI Insights

What is a CTO?

A Chronic Total Occlusion (CTO) is:

- A complete blockage of a coronary artery
- Duration of >3 months
- Responsible for clinically significant decrease in blood flow
- Can be made of hard or soft plaque





What is a microcatheter?

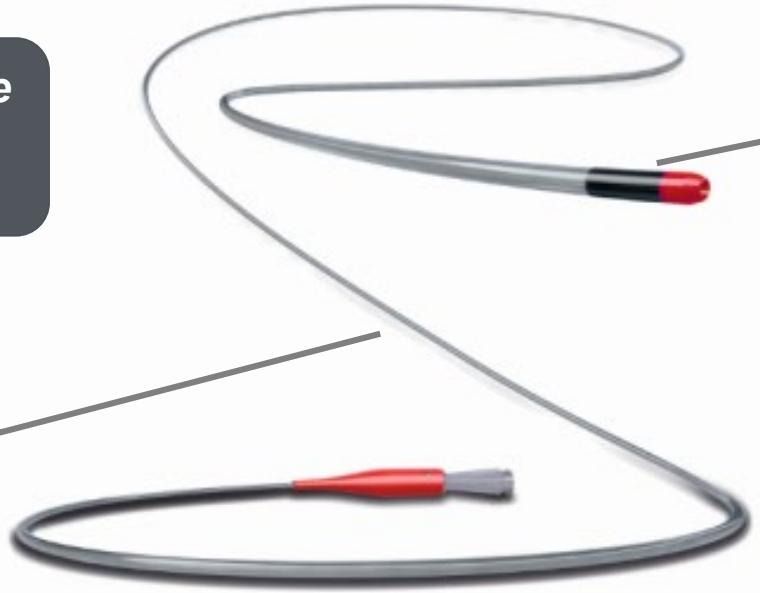
A small-diameter device used to:

- Facilitate wire exchanges
- Maintain wire position between wire exchanges
- Improve wire torque response
- Provide backup support to the coronary guidewire

MAMBA™ Microcatheter *Family Overview*

With their proprietary tapered coil and integrated tip, MAMBA™ torquable microcatheters are engineered to be resilient and optimize support without compromising flexibility and profile.

Small Crossing Profile
MAMBA Flex (2.1F)
MAMBA (2.4F)



Integrated Tip

Resilient, Tapered Coil Shaft

HydroPass™: Durable & Lubricious Coating

MAMBA (135cm)

- ✓ 3 coil taper zones for exceptional wire penetration support
- ✓ Highly pushable and torquable

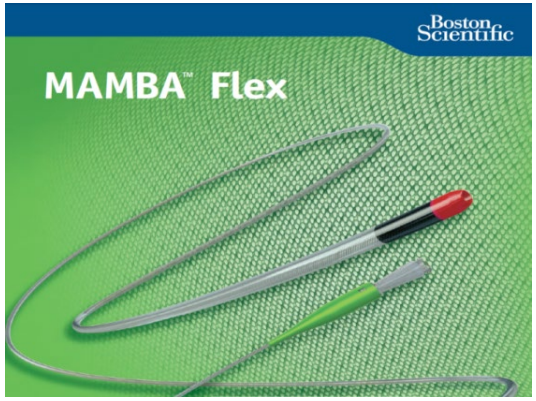
MAMBA Flex (135 & 150cm)

- ✓ 5 coil taper zones for enhanced flexibility, deliverability, and wire follow

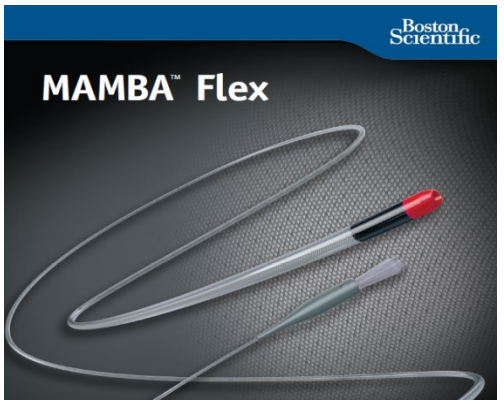
MAMBA™ Microcatheter *Family Overview*



- 135cm
- Low 2.4F Crossing Profile
- 3 taper zones
- Torque Transmission
- Pushability
- Wire Support
- Tip Designed to have support puncture force
- HydroPass™ Coating

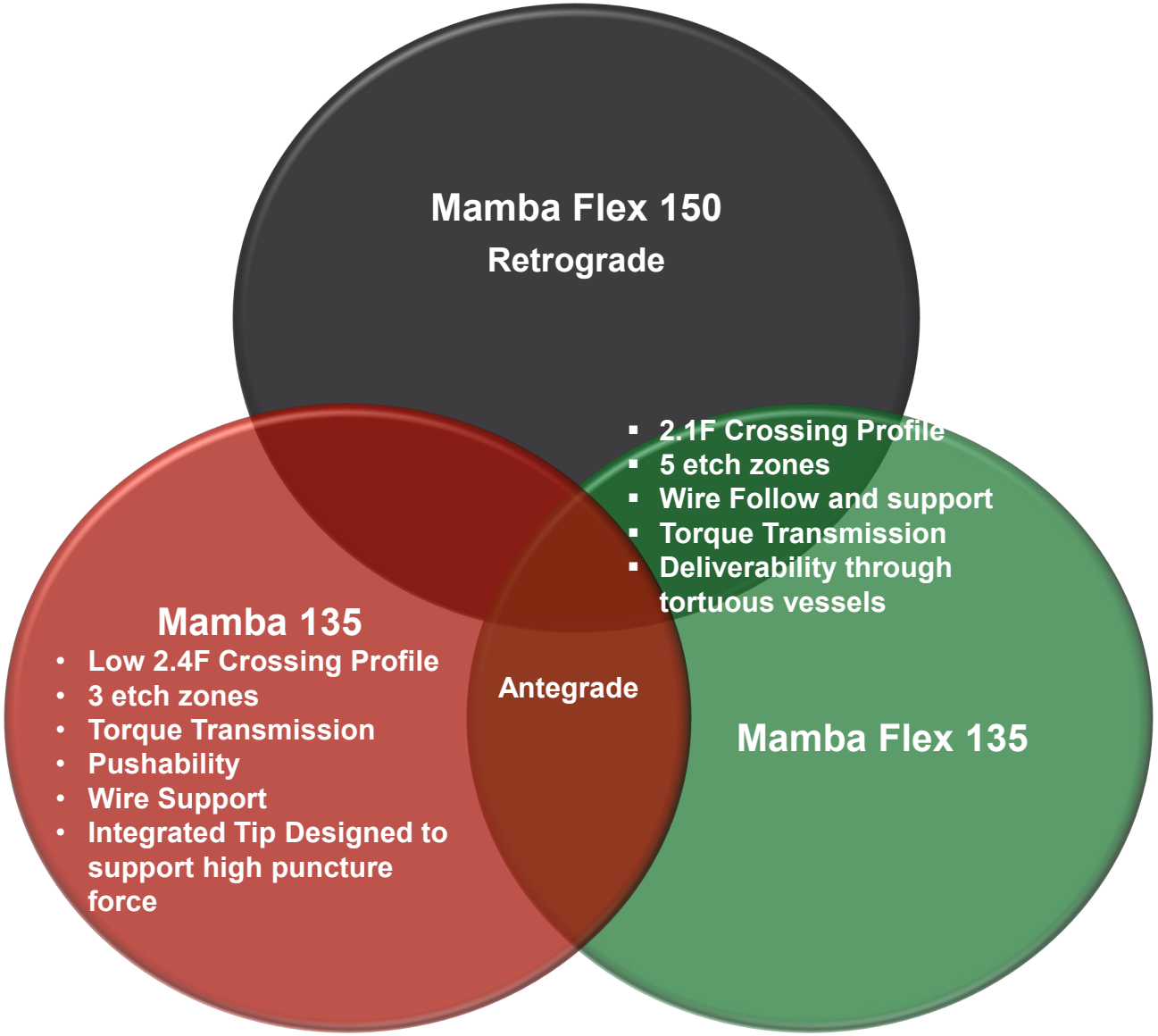


- 135cm
- Low 2.1F Crossing Profile
- 5 taper zones
- Wire Follow and support
- Torque Transmission
- Deliverability through tortuous vessels
- HydroPass Coating



- 150cm
- Low 2.1F Crossing Profile
- 5 taper zones
- Wire Follow and support
- Torque Transmission
- Deliverability through tortuous vessels
- HydroPass Coating
- Added length for retrograde

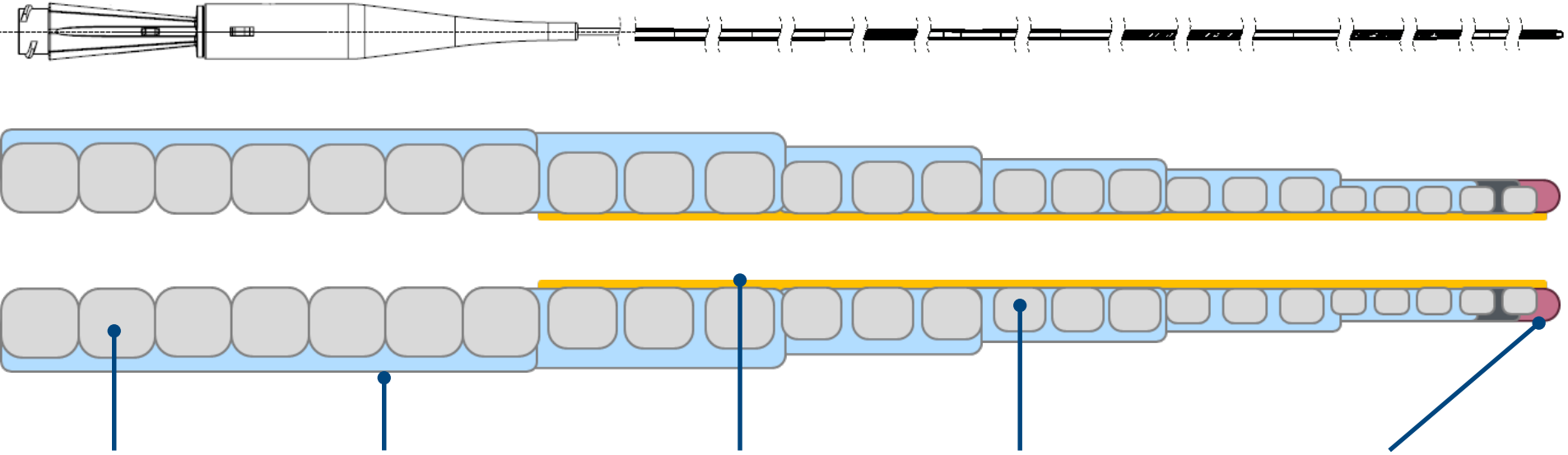
Mamba™ and Mamba™ Flex Microcatheters



MAMBA™ Microcatheter *Design Overview*







Proximal Shaft
- *Resilient*
- *Torqueable*
- *Kink Resistant*
- *Low Profile*

Hydropass™ Hydrophilic Coating
- *Extremely durable & lubricious*

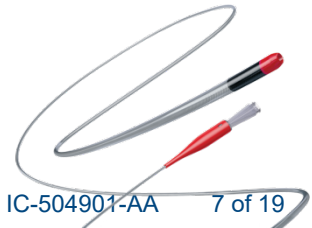
Distal Shaft Liner
- *Reduce wire forces*

Tapered Coil
- *Smooth tapers to minimize weak points*
- *Optimized for desired Flexibility/Support and Length*
- *Low Profile*

Integrated Tip
- *Low profile*
- *Torqueable and Resilient*
- *Coil support extends through the tip*

MAMBA™ Microcatheter *Family Summary*

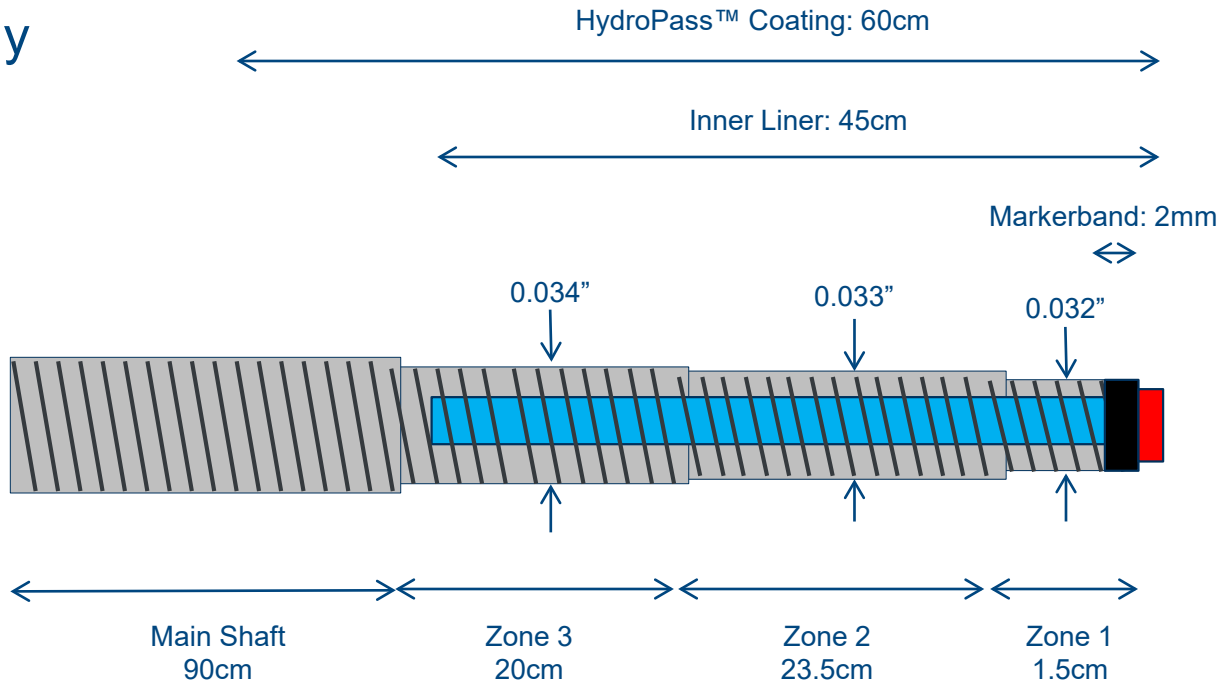
	Marker Band Length	Crossing Profile	Coating	Coating Length (cm)	# of Wires	Tapered Zones	Hub Color	Torqueable?
Mamba	2mm	2.4Fr	HydroPass™	60	11	3	Red	Yes
Mamba Flex 135cm	2mm	2.1Fr	HydroPass	60	11	5	Green	Yes
Mamba Flex 150cm	2mm	2.1Fr	HydroPass	60	11	5	Gray	Yes



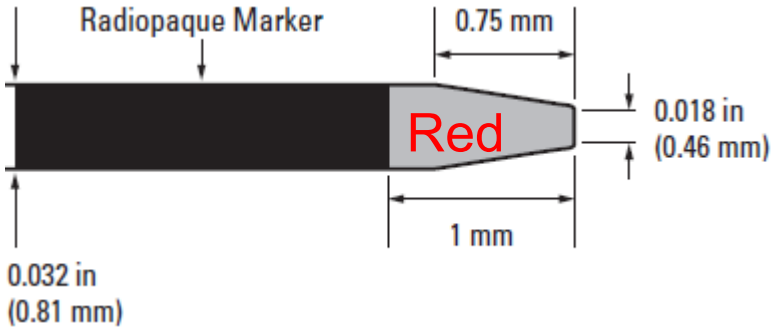
BSC Data on File.

MAMBA™ Microcatheter
135cm Design

Shaft & Body Schematic



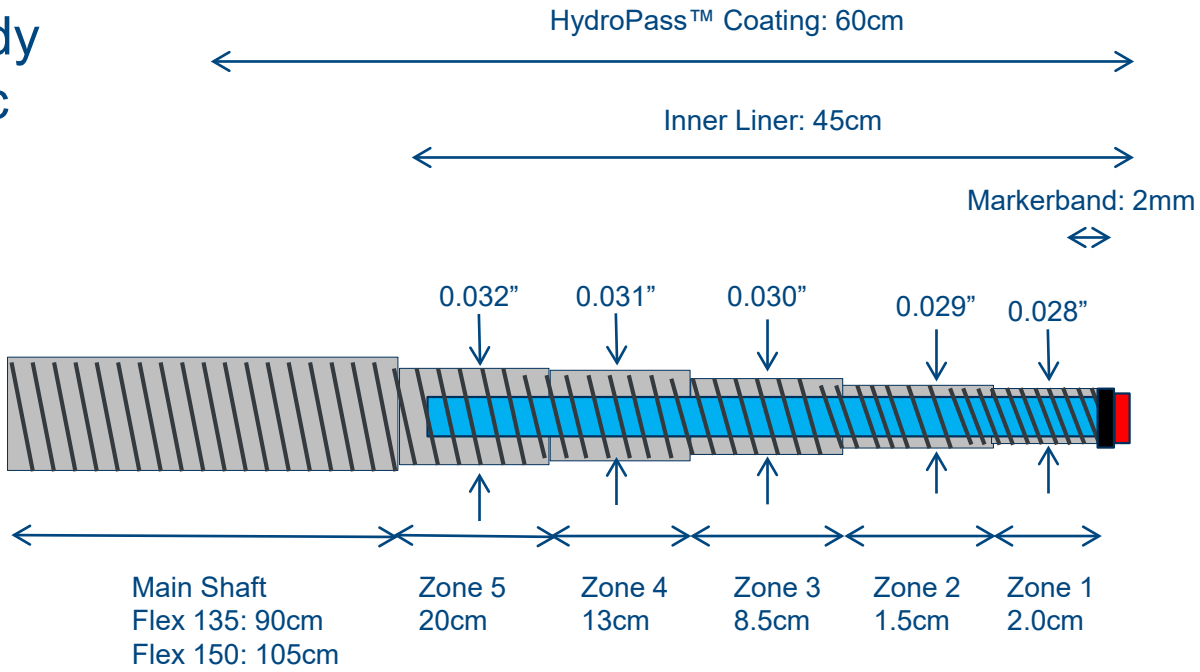
Tip Schematic



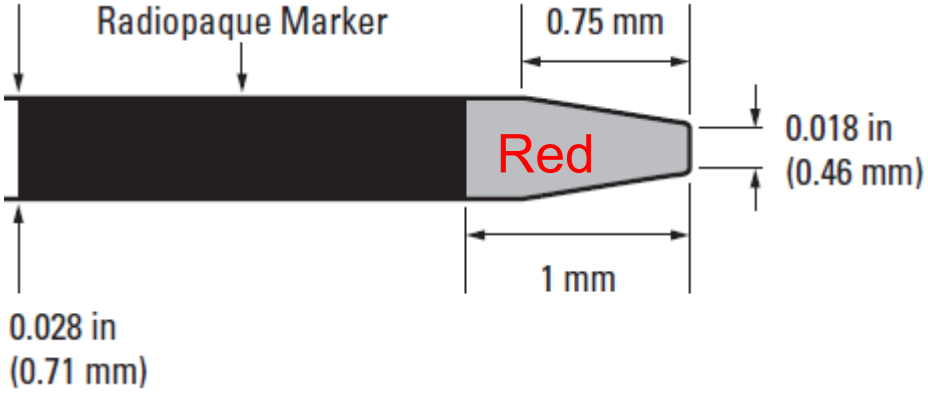
BSC Data on File. MAMBA and MAMBA Flex Microcatheter DFU.

MAMBA™ Flex Microcatheter
135 and 150cm Design

Shaft & Body Schematic



Tip Schematic



Competitive Overview

Boston Scientific

TERUMO	ASAHI	Teleflex Vascular Solutions
FineCross MG Coronary Micro-Guide Catheter	ASAHI Corsair Pro Microcatheter	Turnpike LP catheter
	ASAHI Corsair Microcatheter	Turnpike Spiral catheter
	ASAHI Caravel	Turnpike Gold catheter

Specifications, In Perspective

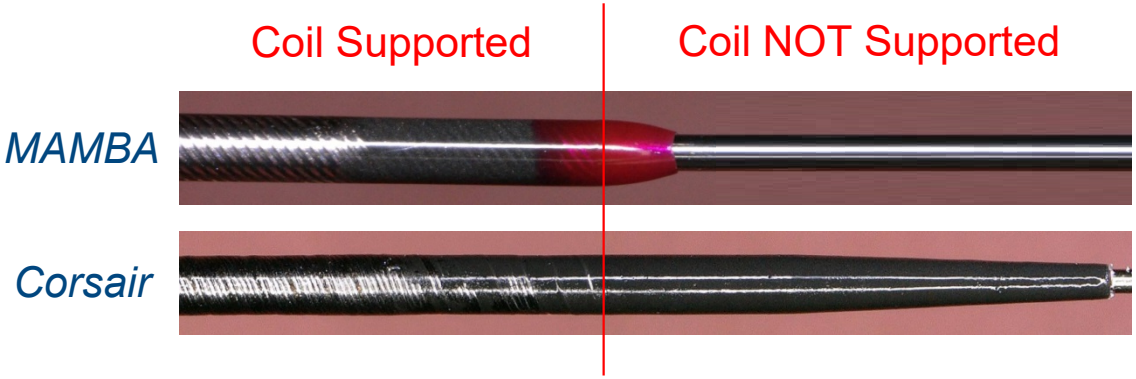
Product	Lesion Entry Profile	Crossing (Distal) Profile	Proximal Shaft Profile	Distal Shaft Inner Diameter	Proximal Shaft Inner Diameter	Lengths available (cm)	Recommended or Maximum GW	Distal Hydrophilic Coating (cm)	Type
Mamba™	0.018”	0.032”	0.038”	0.018”	0.023”	135	≤0.014"		
	0.46 mm	0.81mm	0.97mm	0.46mm	0.58 mm		0.36 mm	60	Coil
	1.4 Fr	2.4 Fr	2.9 Fr						
Mamba Flex	0.018”	0.028”	0.038”	0.018”	0.023”	135/150	≤0.014"		
	0.46 mm	0.71mm	0.97mm	0.46mm	0.58mm		0.36 mm	60	Coil
	1.4 Fr	2.1 Fr	2.9 Fr						
Turnpike™ LP	0.021"	0.029"	0.038"	0.018"	0.018"		≤0.014"		
	0.53 mm	0.74 mm	0.97 mm			135/150	0.36 mm	60	Braid over Coil
	1.6 Fr	2.2 Fr	2.9 Fr						
Corsair™ Pro	0.0165”			0.018”			0.014"		
	0.42 mm	0.87 mm	0.93 mm	0.45 mm	0.45 mm	135/150	0.36 mm	60	Braid over Coil
	1.3 Fr	2.6 Fr	2.8 Fr						
Finecross™	0.021”	0.025”	0.033”	0.018”	0.021”	130/150	0.014”		
		.060mm	0.87mm	0.45mm	0.55mm		0.36mm	70/90	Straw
		1.8Fr	2.6Fr						

MAMBA and MAMBA Flex Microcatheter DFU. Teleflex 2017 Product Catalog.
http://www.terumois.com/products/catheters/finecross.html

MAMBA™ Microcatheter *Design, cont'd*

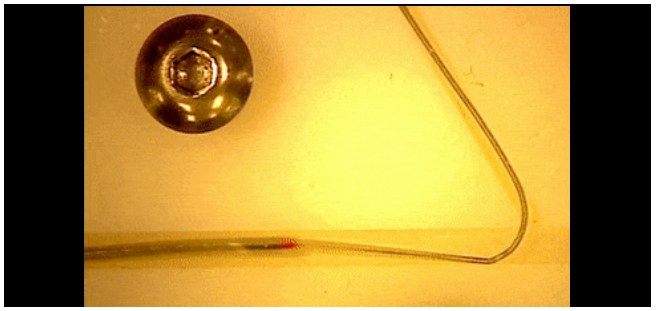
Integrated Tip
Designed to Minimize Failure and Maximize Wire Backup Support

- ✓ Mamba Microcatheter Coil ends less than 1mm from tip
- ✓ Mamba Microcatheter Torque transmission all the way to tip helps deliverability

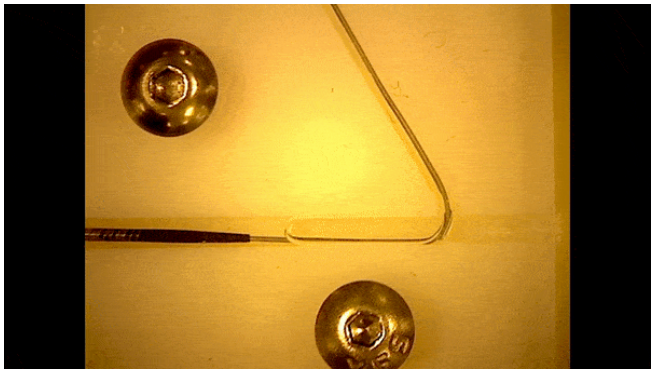


Trackability Performance

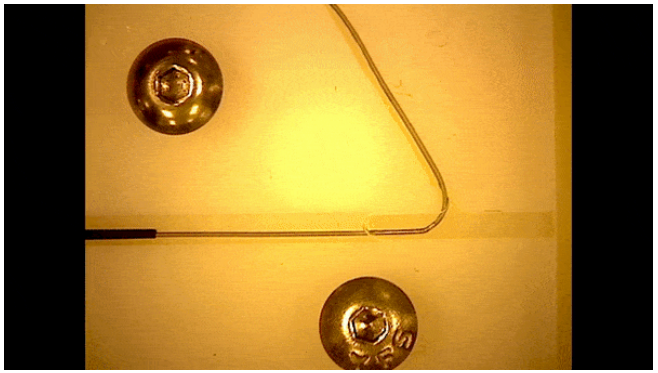
MAMBA™ Flex



Corsair Pro



Turnpike LP



Bench testing performed by Boston Scientific. Results not necessarily indicative of clinical performance. BSC Data on File.

Wire Backup Performance

Boston Scientific

MAMBA™ Flex



MAMBA



Corsair Pro

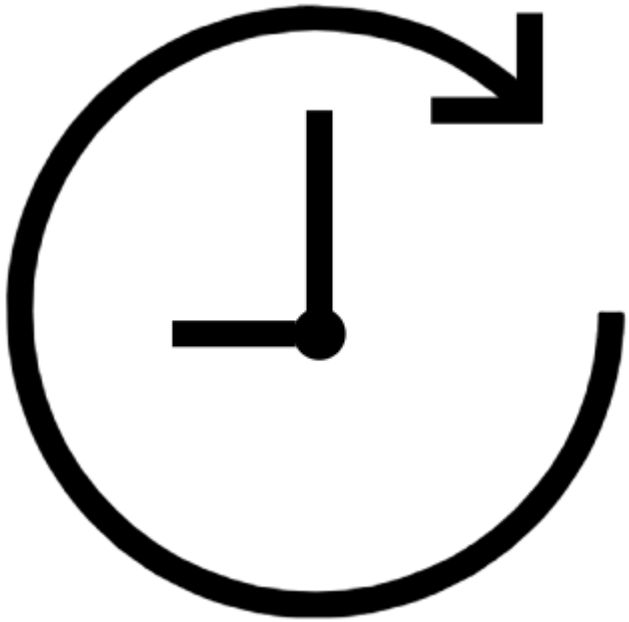


Turnpike LP



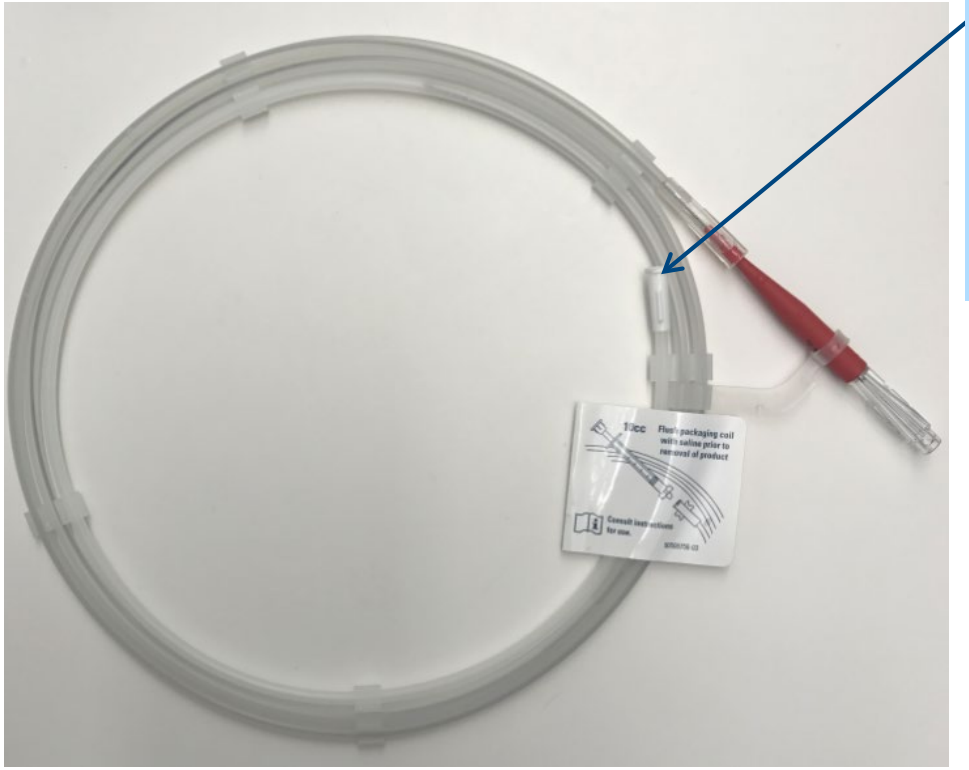
Bench testing performed by Boston Scientific. Results not necessarily indicative of clinical performance. BSC Data on File.

MAMBA™ *Microcatheter Family*



- Advance Mamba Microcatheter with a combination of push and clockwise rotation
- When advancing under resistance, rotate the catheter in either a clockwise or counter-clockwise (but not both) direction while gently pushing
- If the distal tip is not rotating and advancing, do not rotate the microcatheter more than five (5) consecutive rotations in either the clockwise or counter-clockwise direction.

MAMBA™ *Microcatheter Family*



- Before removing the microcatheter from the carrier hoop, flush the carrier hoop with saline with a 10 ml luer-lock syringe to aid in removal and activate the hydrophilic coating.
- Once the product is hydrated, do not allow it to dry.

Mamba™ Microcatheter

Packaging and Ordering

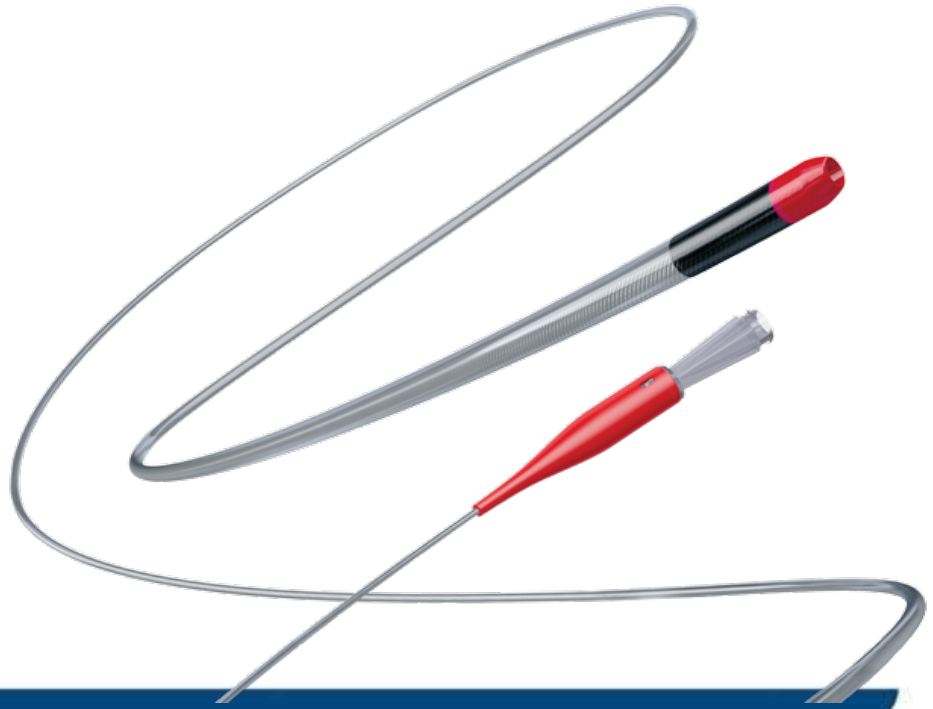






Ref Catalog Number	Material Description	GTIN	GS1 Bar Code with 01 Application Identifier
H7493928713500	MAMBA 135 CM US	08714729940241	
H7493928713540	MAMBA FLEX 135 CM US	08714729940289	
H7493928715000	MAMBA FLEX 150 CM US	08714729940326	

THANK YOU!!



MAMBA™ Microcatheter *Brief Summary*

INTENDED USE/INDICATIONS FOR USE

The MAMBA™ and MAMBA™ Flex Microcatheter is intended to provide support to facilitate the placement of guidewires in the coronary vasculatures, and can be used to exchange one guidewire for another. The microcatheter is also intended to assist in the delivery of contrast media into the coronary vasculatures.

CONTRAINDICATIONS

This device should not be used in neurovasculature.

WARNINGS

- The microcatheter is to be used over a guidewire. Use of the microcatheter ahead of a guidewire may result in device damage and/or vessel trauma.
- Proximal marks should be used to gauge microcatheter position within the guide catheter to avoid over-advancement of the microcatheter.
- Use of this product in narrow or tortuous vessels or over-torqueing the device in challenging anatomy increases risk of vessel trauma including perforation.
- Advancement of the microcatheter under resistance may cause microcatheter and/or vessel damage.
- Advancement of the microcatheter through a stented vessel can damage the microcatheter and/or cause stent displacement.
- If a blocked/damaged lumen is encountered, e.g. when flushing with saline, infusing with contrast or advancing over a guidewire, the microcatheter should be replaced with a new device. Failure to do so may result in catheter rupture, guidewire lockup, or arterial injury.
- It is recommended that this product only be used at a medical institution capable of promptly performing emergency open-heart surgery.

PRECAUTIONS

- Flush and wipe down device with sterilized saline prior to use to ensure smooth movement of guidewire.
- Ensure all appropriate accessories are securely attached to the microcatheter prior to flushing.
- The microcatheter should not be advanced ahead of the guidewire to avoid device damage.
- Always verify tip position before injecting contrast.
- Prior to advancing or withdrawing microcatheter, loosen hemostasis valve to avoid damage.
- Do not over tighten Y-adapter/hemostasis valve to avoid microcatheter damage.
- Ensure removal of guidewire prior to contrast injection.
- Flush microcatheter after contrast injection to avoid guidewire lockup.
- Withdrawing device too quickly or with too much force could cause the device to stretch or the tip to separate.
- This product has not been tested for alcohol or drug delivery.

ADVERSE EVENTS

- Allergic reaction, Arrhythmia, Cardiac tamponade, Cardiogenic shock, Death, Embolism, Fever, Hematoma, Hemorrhage, Infection, Myocardial infarction/ischemia, Pericardial effusion, Radiation exposure, Renal failure, Stroke, Thrombosis, Vasospasm, Vessel trauma, including dissection, perforation, or arteriovenous fistula

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